Design and Development of an E-Commerce REST API with Multimedia Support

B.Tech Project Presentation

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Presentation Outline

1. Introduction
2. Problem Statement
3. Objectives
4. System Architecture
5. Implementation Details
6. Code Implementation
7. System Screenshots
8. Results and Discussion
9. Conclusion and Future Work

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Introduction

- Rapid growth of digital commerce requires scalable backend systems.
- Need for secure and efficient REST APIs.
- Handling multimedia data such as product images.
- Spring Boot enables rapid backend development.

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Problem Statement

- Traditional monolithic systems struggle with scalability.
- Managing product images and relational data together is complex.
- Efficient API architecture is needed to manage product catalogs.

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Objectives

- Build a complete CRUD REST API for e-commerce products.
- Implement image upload and retrieval.
- Develop keyword search using Spring Data JPA.
- Maintain modular layered architecture.

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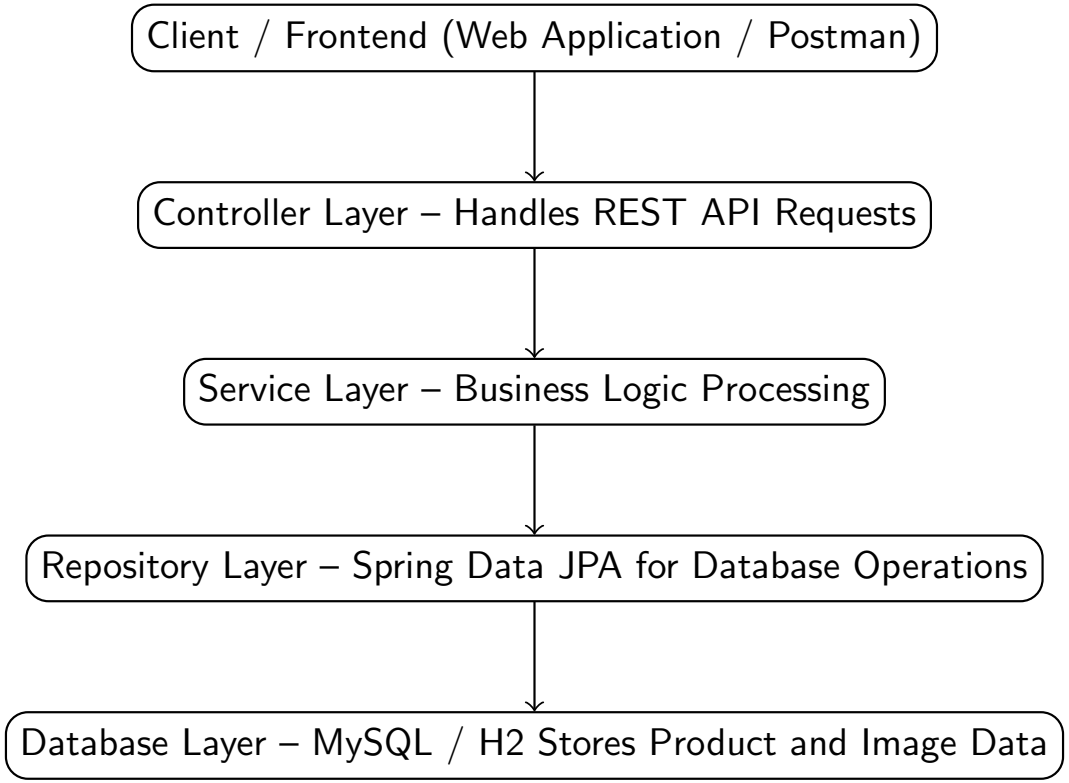
System Architecture



Explanation

- Client sends HTTP requests to the backend system.
- Controller exposes REST endpoints and handles requests.
- Service layer processes application business logic.
- Repository layer communicates with the database.

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Implementation Details

- **Framework:** Spring Boot
- **Language:** Java 21
- **Database:** MySQL / H2
- **Technologies Used:**
  - Spring Web
  - Spring Data JPA
  - Multipart File Upload
  - RESTful API Design

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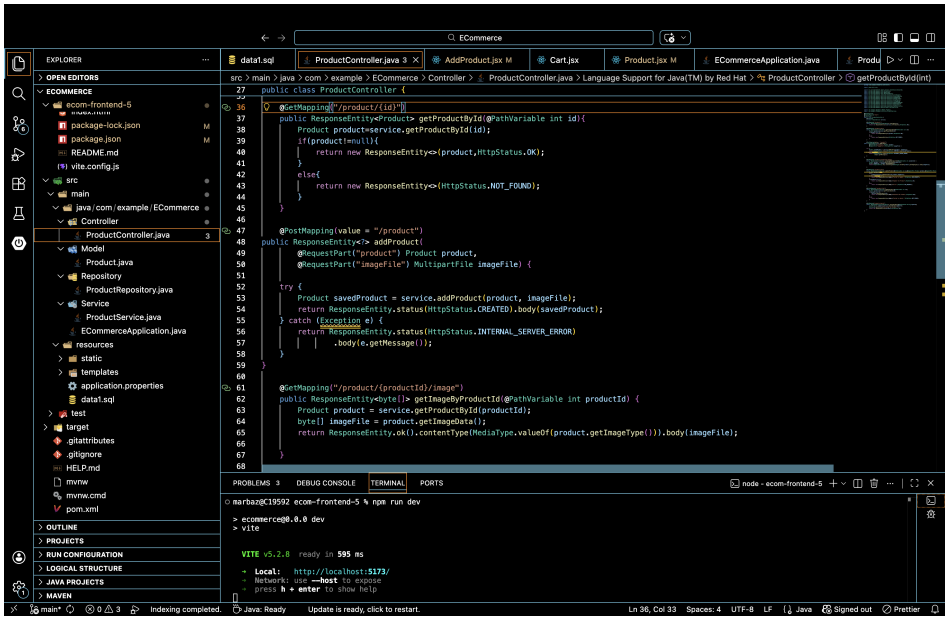
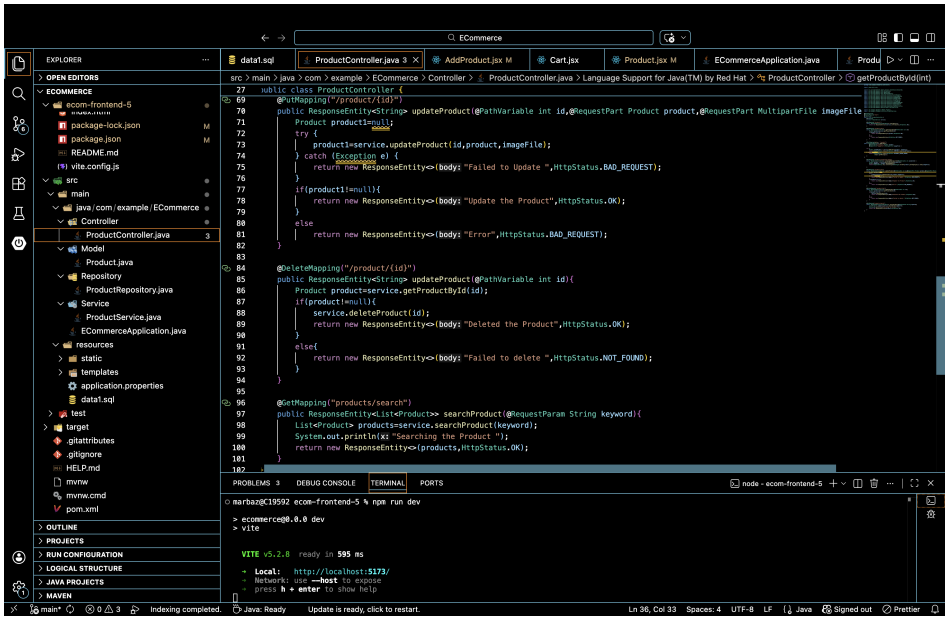
Code Implementation

Controller Code



Service / Logic Code



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API Testing

Fetching All Products (GET Request)

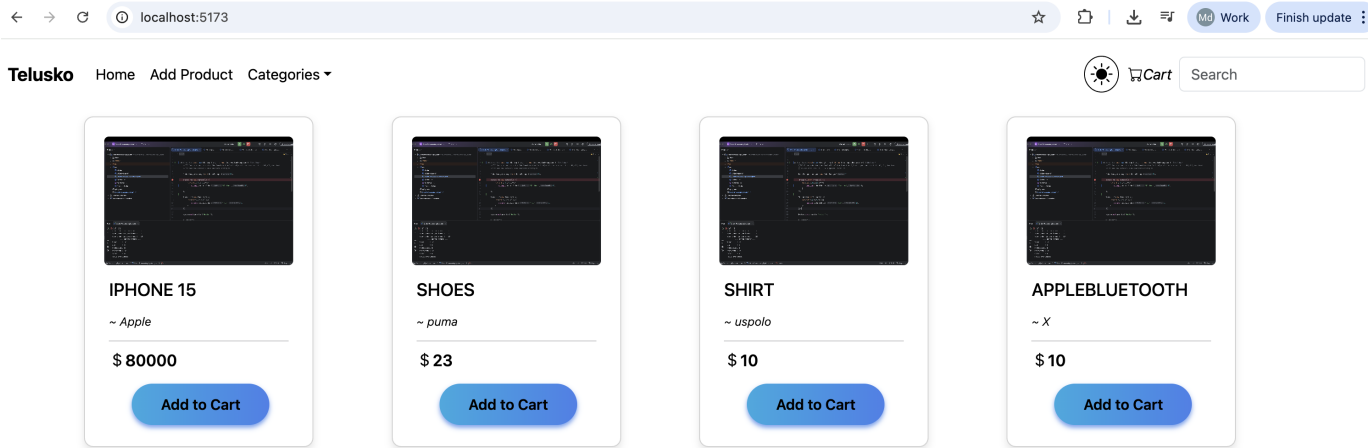
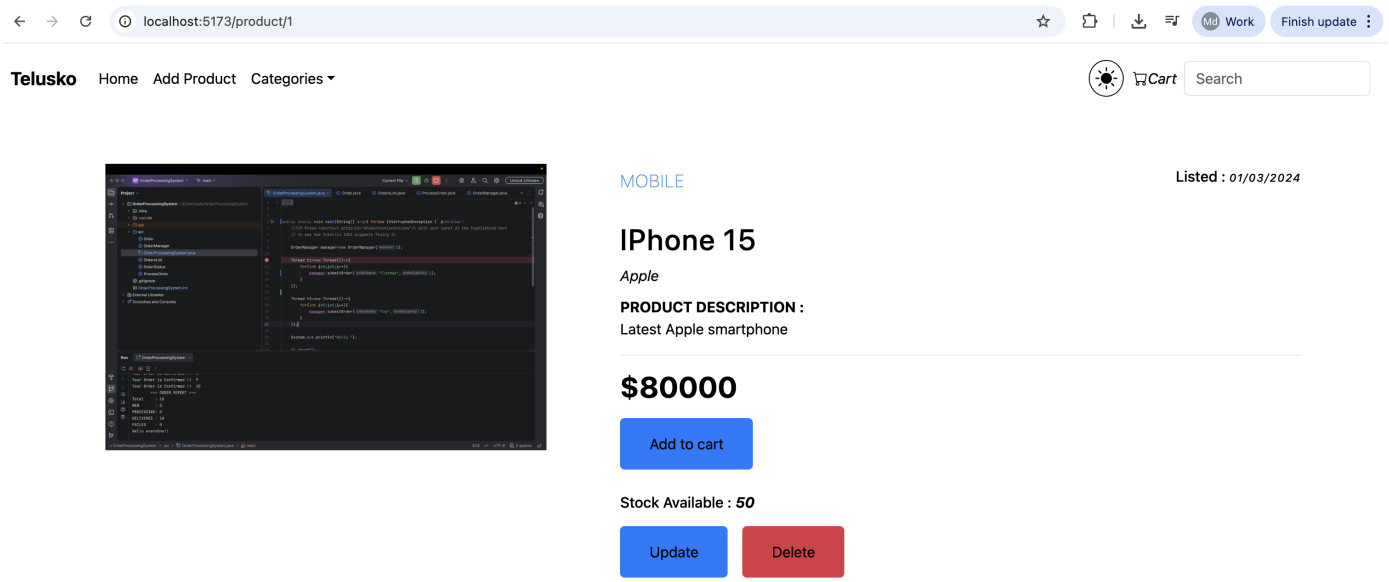


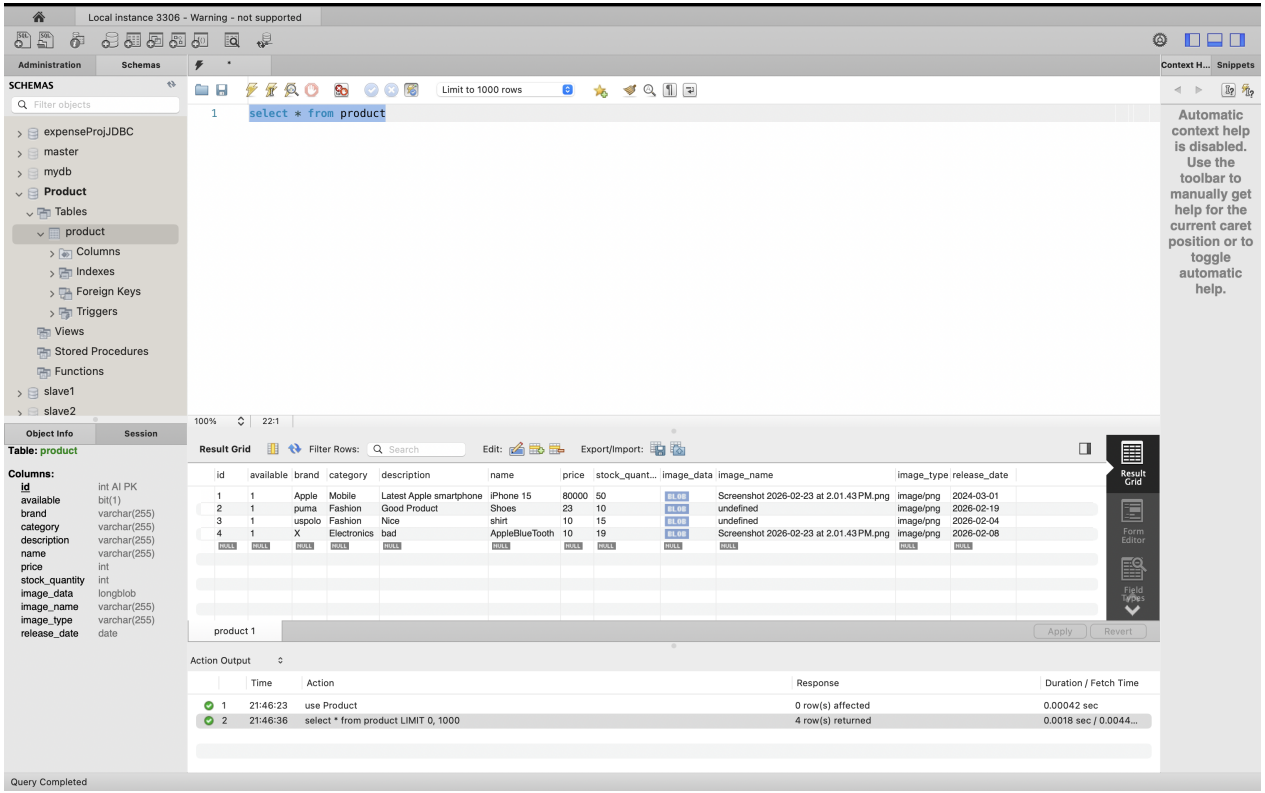
Image Uploading

Adding Product with Image (POST Request)



Database Records

Stored Product Data in Database



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Results and Discussion

- REST API successfully processes JSON and multipart requests.
- Proper HTTP status codes verified (200, 201, 404).
- Images successfully stored as BLOB data in database.
- API tested successfully using Postman.

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Conclusion

- Successfully developed scalable backend for e-commerce systems.
- Implemented REST API with multimedia support.
- Layered architecture ensures modular design.

Future Work

- Implement Spring Security with JWT authentication.
- Convert architecture to microservices.
- Integrate AWS S3 for image storage.

Thank You